$\boxed{\therefore \forall x P(x) \vee \exists x Q(x) \equiv \forall x \exists y (P(x) \vee Q(y))}$

A statement is in $\boxed{\text{prenex normal form (PNF)}}$ if and only if it is of the form $Q_1 x_1 Q_2 x_2 \ldots\ldots Q_k x_k P(x_1, x_2, \ldots, x_k)$, where each Q_i, $i = 1, 2, \ldots, k$, is either the existential quantifier or the universal quantifier, and $P(x_1, \ldots, x_k)$ is a predicate involving no quantifiers. For example, $\exists x \forall y (P(x,y) \wedge Q(y))$ is in prenex normal form, whereas $\exists x P(x) \vee \forall x Q(x)$ is not (because the quantifiers do not all occur first).

50) Put these statements in prenex normal form.

a) $\exists x P(x) \vee \exists x Q(x) \vee A$, where A is a proposition not involving any quantifiers.

$\exists x P(x) \vee \exists x Q(x) \vee A \equiv \exists x P(x) \vee \exists x (Q(x) \vee A)$ (By sec 1.4 Ex-48(b)).

$\equiv \exists x (P(x) \vee (Q(x) \vee A))$ (By Sec 1.4, Ex-47)

b) $\neg(\forall x P(x) \vee \forall x Q(x))$

$\neg(\forall x P(x) \vee \forall x Q(x)) \equiv \neg \forall x P(x) \wedge \neg \forall x Q(x)$ (By 2nd De Morgan's Law) $\equiv \exists x \neg P(x) \wedge \exists x \neg Q(x)$ (By De Morgan's law of qu

b) $\neg(\forall x P(x) \vee \forall x Q(x))$

$\neg(\forall x P(x) \vee \forall x Q(x)) \equiv \neg(\forall x \forall y (P(x) \vee Q(y)))$ (Ex-1.5, Prob 48)

$\equiv \exists x \neg \forall y (P(x) \vee Q(y))$ (De Morgan's law for quantifiers)

$\equiv \exists x \exists y \neg (P(x) \vee Q(y))$ (De Morgan's law for quantifiers)

$\equiv \exists x \exists y (\neg P(x) \wedge \neg Q(y))$ (2nd De Morgan's Law)

c) $\exists x P(x) \to \exists x Q(x)$

$\exists x P(x) \to \exists x Q(x) \equiv \neg \exists x P(x) \vee \exists x Q(x)$ (Conditional-disjunction equivalence)

$\equiv \forall x \neg P(x) \vee \exists x Q(x)$ (De Morgan's law for quantifiers)